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BRAIN SPECIALIZATION FOR MUSIC AND WORD READING: A HIGH-RESOLUTION PRECISION IMAGING STUDY

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INTRODUCTION

Litterature:

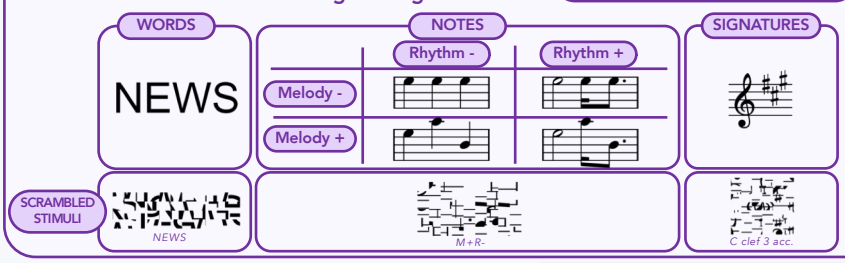
- With literacy, parts of the Ventral Occipito-Temporal Cortex(VOTC) become specialized for the recognition of strings of letters or numbers [1, 2].
- High-resolution 7T MRI imaging further reveals fine-grained specialization for different scripts [3].
- In the field of music, behavioral and neuropsychological dissociations exist between the processing of melody and rhythm[4].

Questions:

- Are there distinct patterns of specialization for reading words and music in the visual cortex?
- Within music, are there distinct patterns of specialization for notes and for other musical symbols such as flats or clefs?
- How sensitive is the visual region specializing in musical symbols to melody and rhythm?
- What is the influence of musical expertise on specialization?

METHODS

Image categories:



Participants: 17 highly trained musicians and 19 controls musically illiterate controls, all right-handed

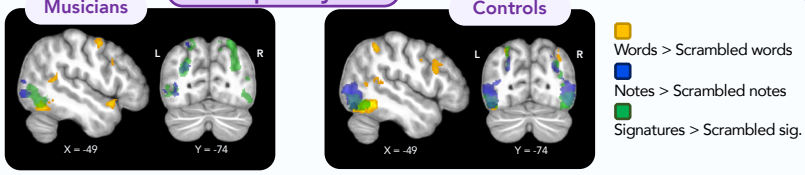
Acquisition: 7T MRI
(block design, whole brain, resolution 1.5 mm)

Oddball detection task:

RESULTS

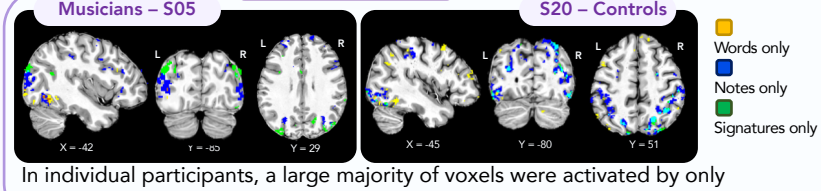
Specialization for different symbols

Group analyses

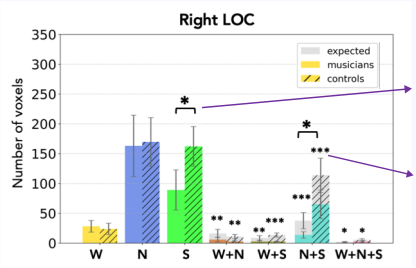


Partially distinct activations for W, N and S. No group difference.

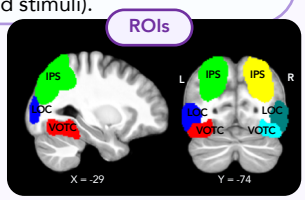
Individual analyses



In individual participants, a large majority of voxels were activated by only one type of symbol (after subtraction of scrambled stimuli).



An example ROI: the right Lateral Occipital Cortex (LOC)



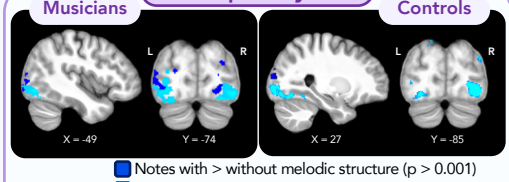
Fewer voxels are activated by signature only in control than in musicians.

Fewer voxels are activated by both notes and signature than predicted if both features were independent, indicating specialization for diffent types of symbols.

In some ROIs, this fine-grained specialization was increased in musicians relative to controls.

Coding of melody & rhythm

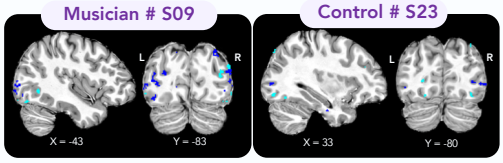
Group analyses



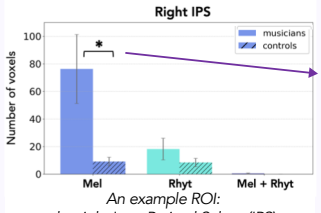
Notes with > without melodic structure (p > 0.001)
Notes with > without rhythmic structure

Dissociation between Mel and Rhyt activations. No group difference.

Individual analyses



Different voxels were sensitive to melody and to rhythm.



An example ROI: the right Intra Parietal Sulcus (IPS)

More voxels are sensitive to melody in musicians than controls.

DISCUSSION

Distinct activations for music and for words:

- **Words** activate mostly the Visual Word Form Area in the "What" visual pathway (VOTC)
- **Notes of music**, whose pitch is represented by position in space, activate mostly the "Where" pathway (LOC, IPS)
- **Other musical symbols**, which are visually akin to letters, but whose role is to give meaning to the position of notes, activate both pathways
- Also illustrating functional specialization, different regions are sensitive to **melodic and to rhythmic complexity**

Fine-grained specialization for different symbols:

- Even in regions in which group-level analyses show an overlap, individual analyses show that voxels sensitive to more than one single type of symbol are a rare exception
- At the current stage of the analyses, musical expertise appears (1) to reduce the number of voxels activated by musical symbols, and (2) to increase the degree of voxel-level specialization.

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